“Empowering Science and Data Analytics through Python”

Assignment For

Lecture_01

Topic:

1. **Operators and Expressions,**
2. **Arithmetic/logic operators,**
3. **Type conversion,**
4. **F-strings,**
5. **Intro to debugging.**

Instructions:

Practice without using pc to check your knowledge.

1. **Operators and Expressions**
   1. **Identify expression or statement**
   i) x + y * 2
   ii) a = b + c
   iii) 5 > 3
   iv) Total = price * quantity

   2. **Find the output:**
      i) a = 10
         b = 3
         print(a + b, a – b, a*b, a/b, a //b, a%b)

      ii) 15 % 4 + 10 //3

   3. **Find out:**
      i) 5 + 2 * 3 and add bracket so that the result becomes 21.
      ii) Evaluate: 10 – 4*2 + 6/3
      iii) 10 + 5 * 2 ** 3 // 4 - 6 % 4 + (3 * (2 + 1)) and not (5 > 3 and 2 < 1)

   4. **Write Expression:**
      Add two numbers and divide by 2

   5. **Write an expression to :**
      Check if a number is even

   6. **Write an expression to :**
      Find square of a number

2. **Logical Operators:**

   Practice:

   1. Find Output:
      i) print(True and True)
      ii) print(False and True)
      iii) print(not False)

   2. Evaluate
      i) print((10 > 5) and (3 == 3))
      ii) print((4!=4) or (6 > 2))
      iii) print(not (7 <=7))

3. Find the output:
   i) print(0 and 5)
   ii) print(10 or 0)
   iii) print(“” or “Hello”)
   iv) print(5 >3 or 10/0)
   v) print(2 <1 and 10/0)

4. Evaluate:
   i) print(None and 5)
   ii) print(5 and None)
   iii) (5 and 0) or (3 and 4) and not (2 >1)

5. Write a logical expression:
   Check if a number is between 10 and 20.

6. Write:
   Check if a number is NOT equal to 5.

3. Comparison Operators:

Practice:

1. Evaluate:
   i) 10 > 5 + 3
   ii) 7 * 2 == 14
   iii) 15 /3 < 4
   iv) 9 % 2 == 1

2. Compare:
   i) 5 + 2 > 3 * 2
   ii) 12 – 4 <= 2*4

3. True or False
   i) 100 > 50*2
   ii) 20 / 5 == 2 + 2
   iii) 3**2 < 10

4. Evaluate
   i) (8+2)> (3*3)
   ii) (6/2) == (9 -6)
   iii) 5 == “5”

5. Fill the correct operators
   i) 7 --- 7
   ii) 9 --- 10
   iii) 5 --- 3
   iv) 4 --- 4

4. Type Conversion:

Practice:

1. Find out
   i) int(5.9)
   ii) float(10)
   iii) str(25)
   iv) int(“10”) + 5
   v) float(“3.14” + 2
   vi) int(“5.9”)

2. Find output:
   i) int(True) + int(False)
   ii) str(10) + str(20)
   iii) int(False), float(True)

3. Find the output:
   i) int(“abc”)
   ii) bool(0), bool(1), bool(-5)
   iii) bool(“”), bool(“Hello”)

4. Find output:
   i) int(3.7) + float(“2.3”) + int(True)
   ii) str(int(“5”) + float(“2.0”)
   iii) bool(int(“0”)) + int(bool(“False”)

5. Which will give error?
   i) int(“0”)
   ii) int(“10.5”)
   iii) float(“10.5”)
   iv) str(10.5)

5. f- Strings

Practice:

1. Find the output:
   i) name = “Shyam”
      print(f “Hello {name}”
   ii) a =5,
      b= 3
      print(f “{a} + {b} = {a +b}”)
   iii) x = 10
      print(f “Value of x is {x*2}”
   iv) name = “John”
      print(f “{name} is {age} years old”

v) print(f “{10/3:.2f}”

2. Find the output:
   x = 10
   print(f “Value of x is {x*2}”

3. Evaluate:
   name = “John”
   age = 25
   print(f “{name} is {age} years old”)

4. What is the output?
   a = 5
   print(f “{a**2}{a +10}”)

5. Find the output:
   i) print(f “{True}{False}”)
   ii) num =7
   print(f “{num} is {‘even’ if num%2 == 0 else ‘odd’}”
   iii) print(f “{int(‘5’) + float(‘2.5’)}”)
   iv) value = 3.14159
   print(f “{value: .3f}”)
   v) a = “10”
   b = 5
   print(f “{int(a) + b}”

6. Operator Precedence

Practice:

1. evaluat:
   i) 5 + 2*3
   ii) 10 – 4/2
   iii) 6 + 3*2
   iv) 8*2 5
   v) 20/5 + 3
   vi) 10 + 2*3**2
   vii) 18/3*2
   viii) 7 + 6/3**2
   ix) 5 + 2*3 – 4
   x) 10 – 2 + 3*4

2. Evaluate
   i) 10 + 5*2**2
   ii) 15%4 + 6*2
   iii) 20//3 + 2*3
   iv) 2**3**2
   v) 100/10*2 + 5

vi) 10 + 5*2**3//4 – 6%4
vii) 5 + 3*4 + 8/2**2

3. Add brackets to change result:
   i) 5 + 2*3 → make answer 21.
   ii) 10 – 2*3 → make answer 24.

Debugging

Practice:

1. What is wrong?
```
a = 5
b = 2
print(a+b
```

2. Find error:
```
x = 10
y = 5
print(x – y))
```

3. Debug this:
```
num = “10”
print(num + 5
```

4. Find mistake:
```
a = 5
b = 0
Print(a/b)
```

5. Fix error;
```
value = 7
if value =7:
        print(“equal”)
```

6. What is the problem?
```
num = 4
if num % 2 ==1:
        print(“even”)
else:
        print(“odd”)
```